

Instructions



**Tell me if you have problems with
practictical tasks**

How is it???



good
goood

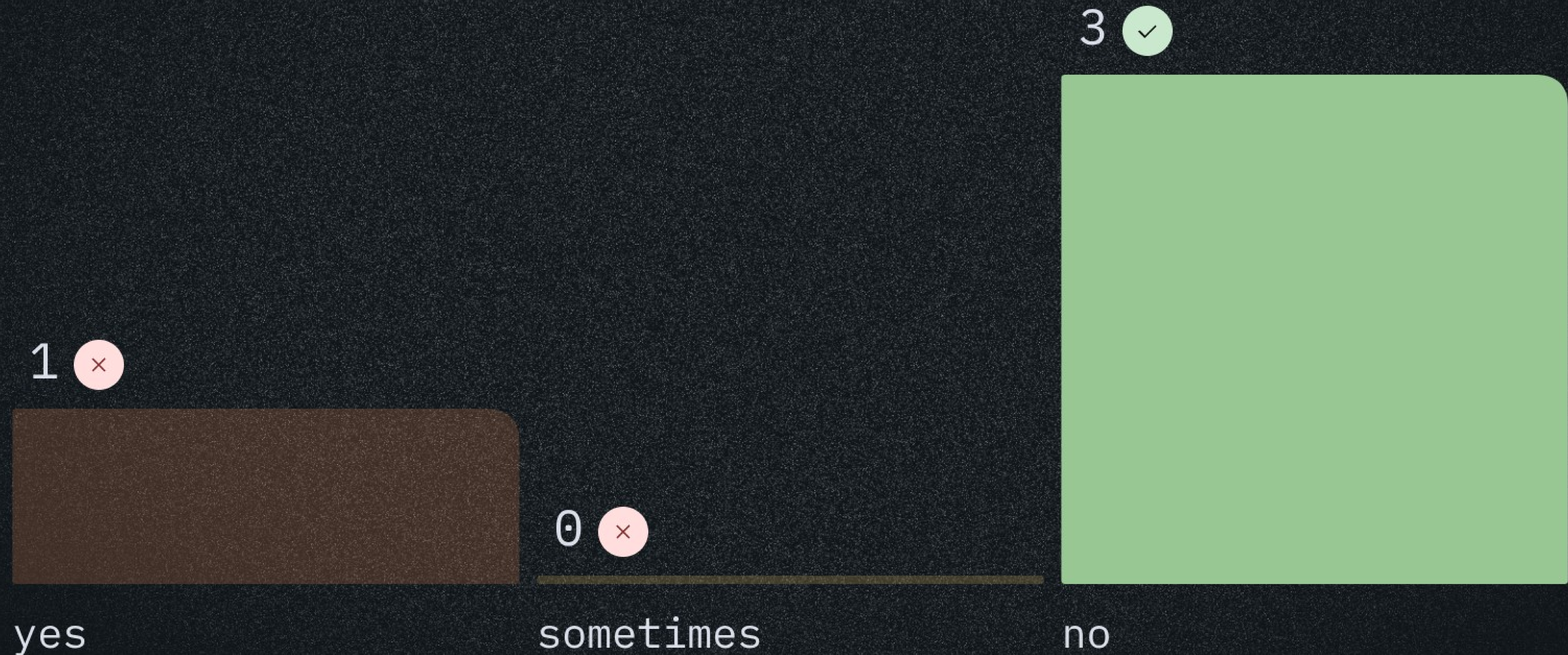
Agenda

- Recall what was learned
- Look at current topic practice
- Discuss theory
- Quiz

Total Recall



Is shallow copy allowed for immutable object?

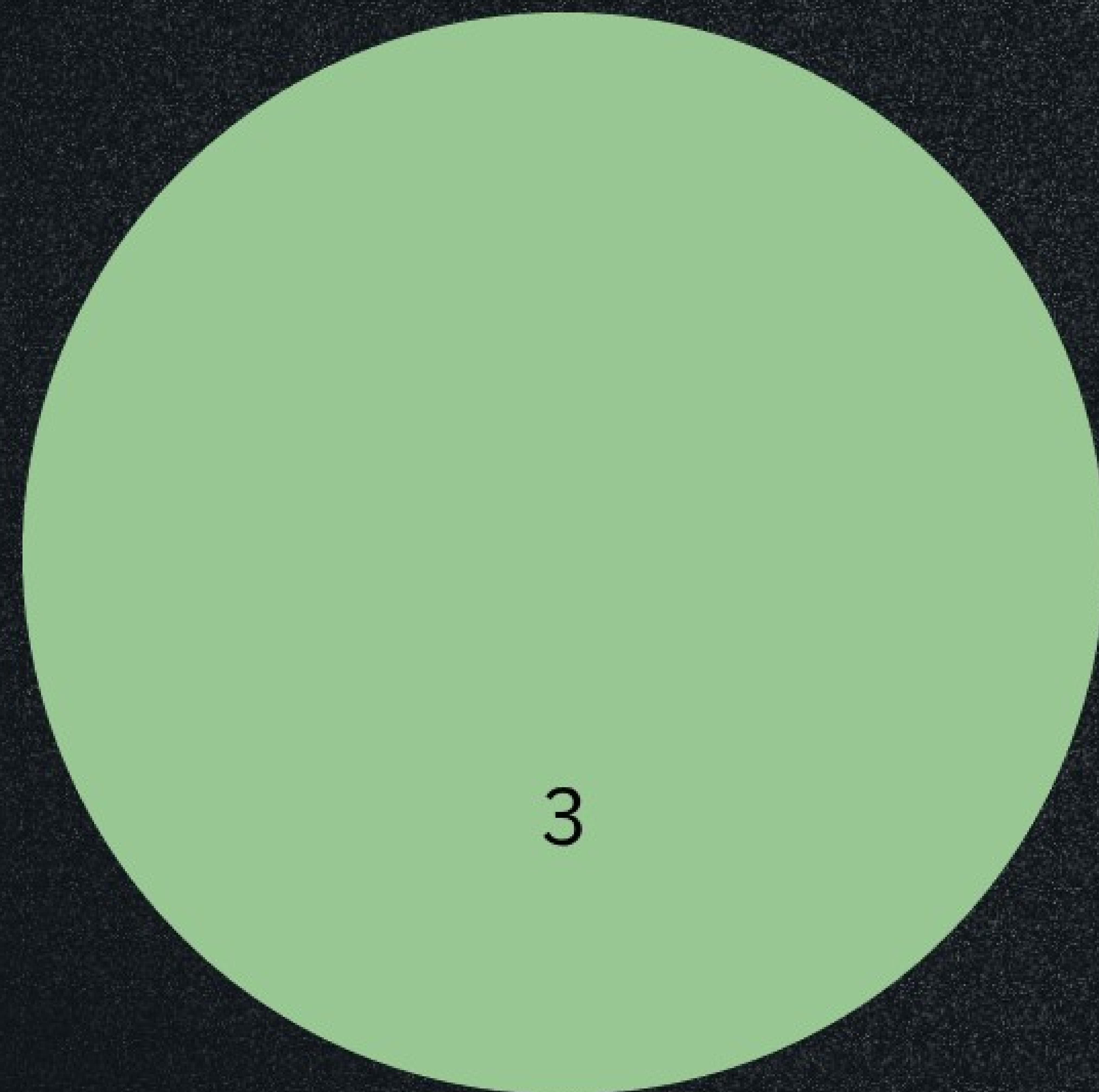


In the Observer pattern, what does the observer implement to receive state changes?



0	attach(observer)	✗
0	detach(observer)	✗
0	notify()	✗
4	update()	✓

How does the WeatherData class (Publisher) notify observers of a state change in the Observer pattern?



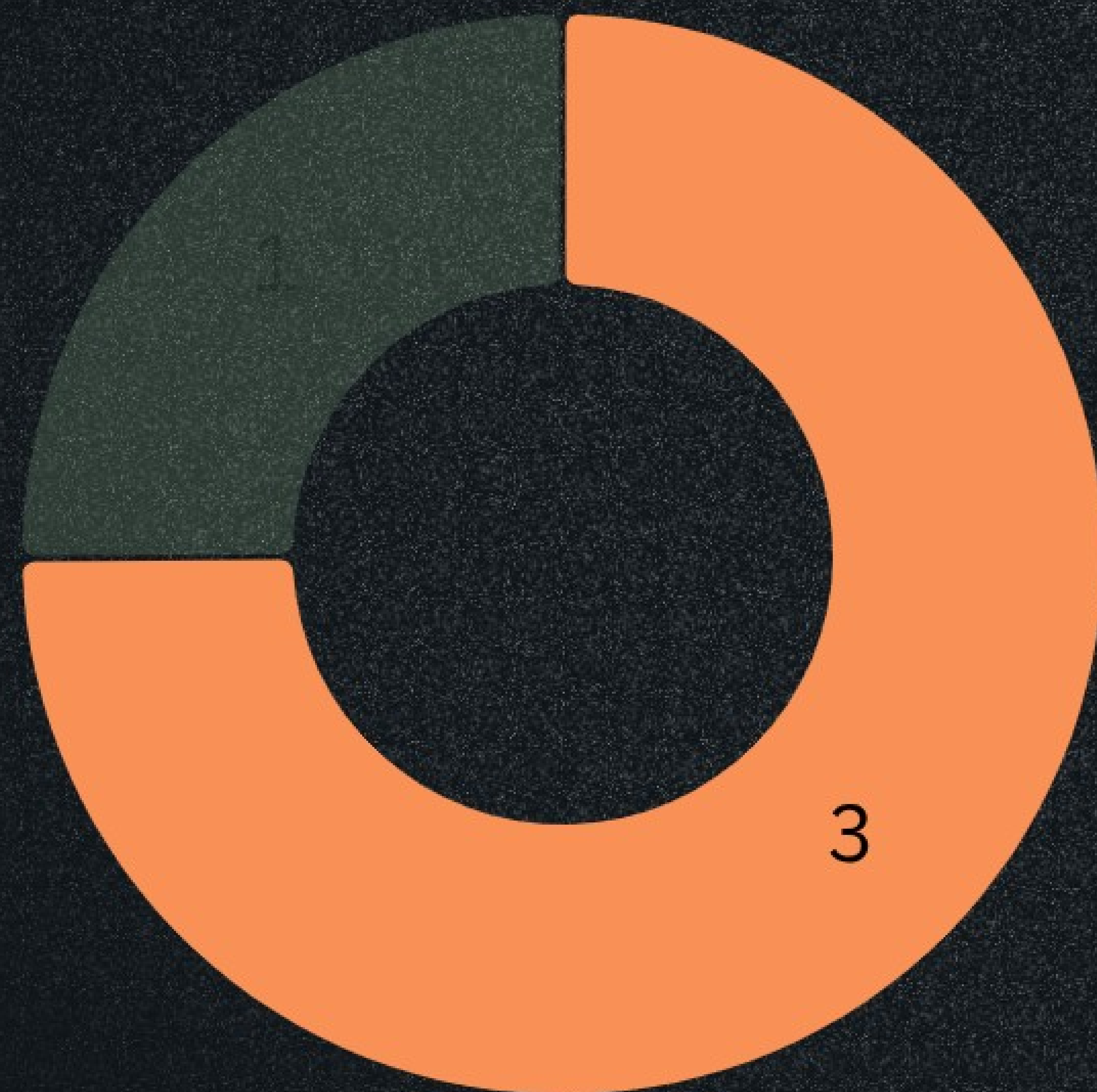
- | | | |
|---|---|---|
| 0 | By triggering an event in the observer objects | ✗ |
| 0 | By directly updating observer objects | ✗ |
| 3 | By calling notifyObservers method and iterating through observers | ✓ |
| 0 | By sending notifications via a separate channel | ✗ |

What is the purpose of the Builder Pattern?



- ☐ 0 To convert the interface of a class into another interface ☐
- ☐ 0 To deposit a specific amount into an account ☐
- ☒ 3 To allow the construction of complex objects step by step and produce different types and representations of an object using same construction code ☒
- ☐ 0 To execute and undo a deposit operation ☐

How does the Builder design pattern achieve better maintainability compared to having numerous constructors with varying parameters?



- | | | |
|---|---|---|
| 3 | a) By allowing the step-by-step construction of complex objects | ✓ |
| 0 | b) By providing a common interface for object construction | ✗ |
| 1 | c) By consolidating all variations of object construction into a single builder | ✗ |
| 0 | d) By incorporating all parameters into a single constructor call | ✗ |



Discuss current topic

The Moon is drifting away from the Earth

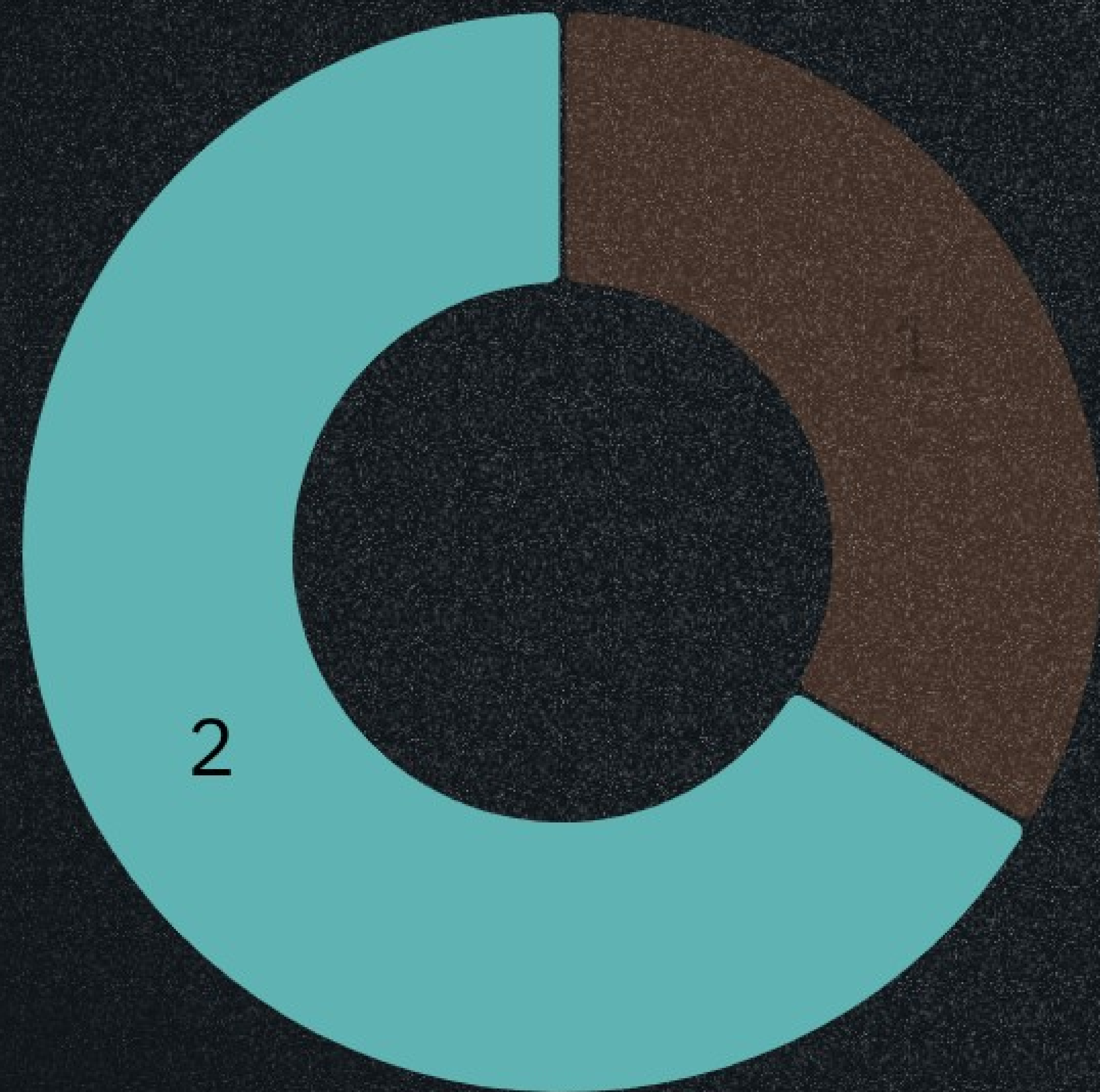
approximately 3.8 cm away from our planet every year.

Have you looked through the materials?



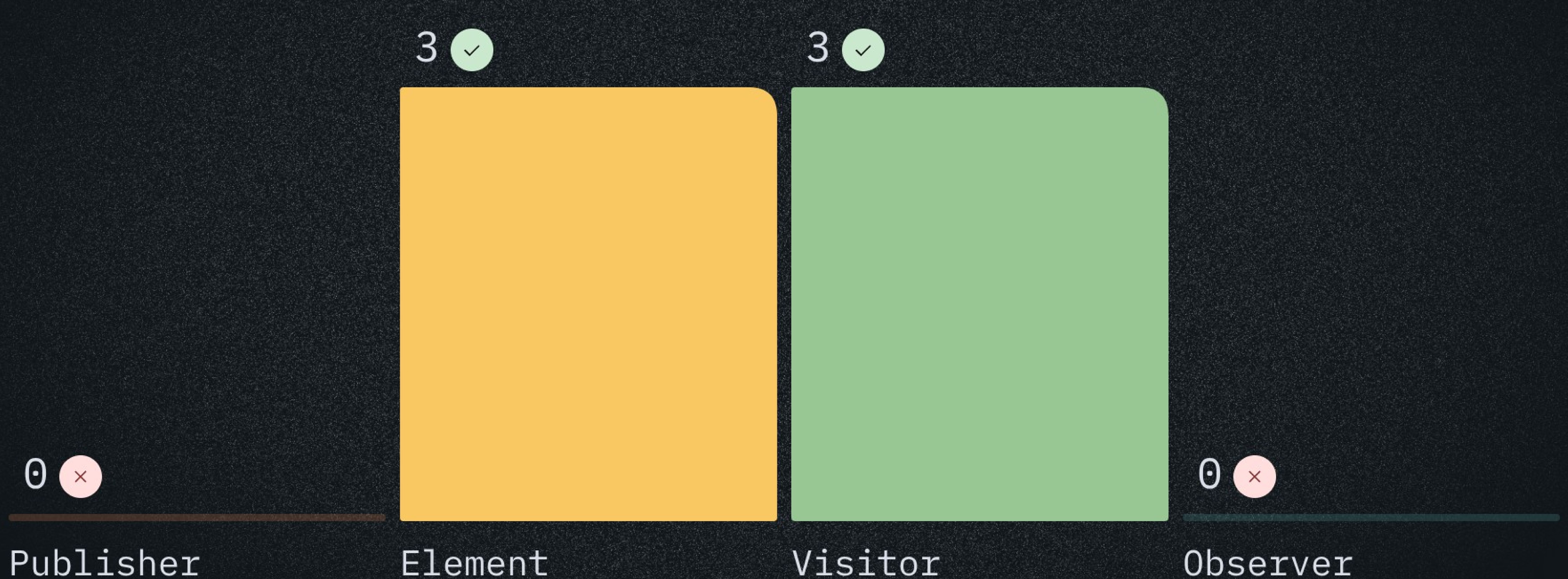
Convinient Patterns: Visitor

When to use Visitor pattern?



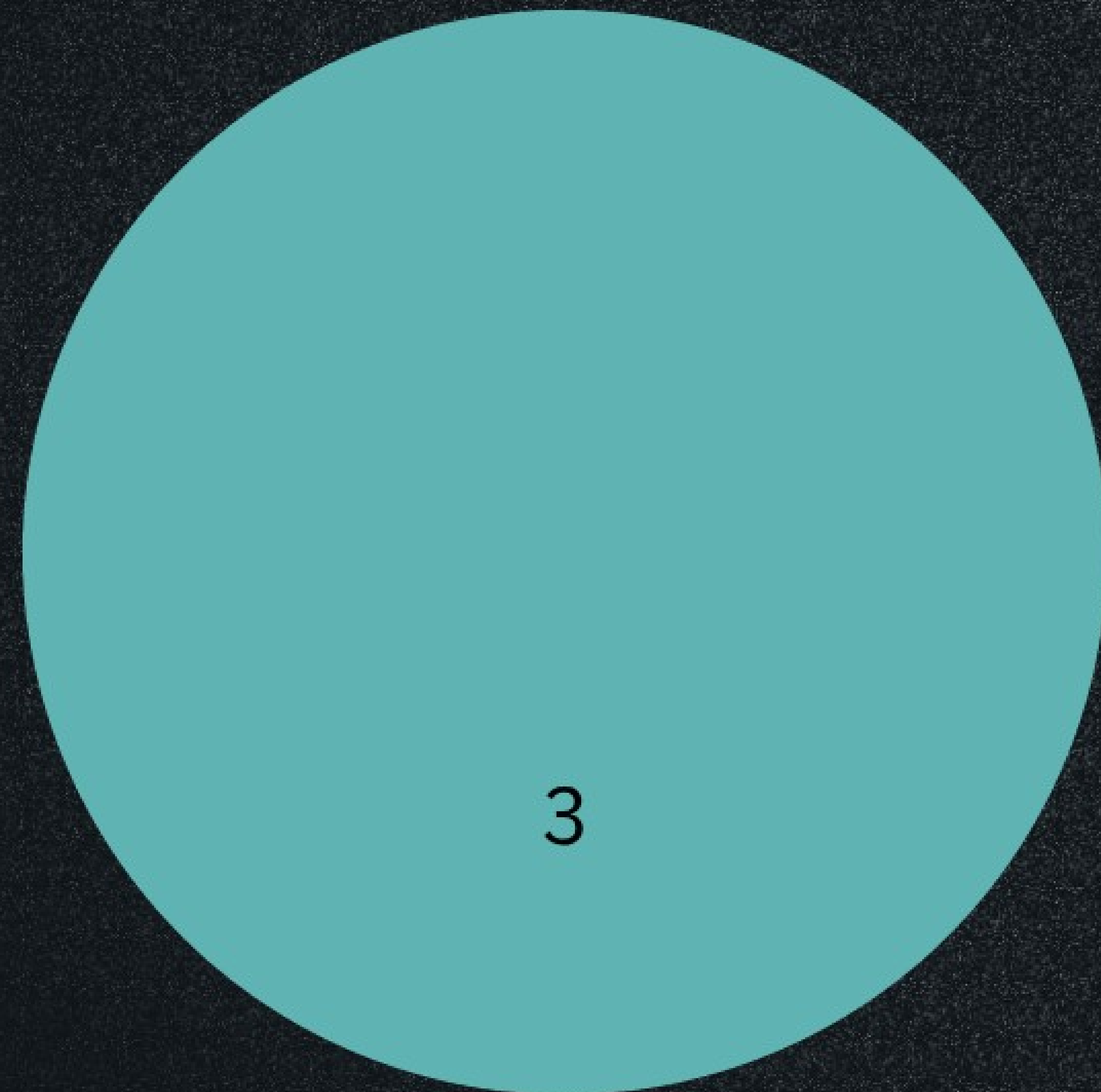
- | | | |
|---|--|---|
| 1 | Use the Visitor when you need to perform an operation on all elements of a complex object structure (for example, an object tree). | ✗ |
| 0 | Use the Visitor to clean up the business logic of auxiliary behaviors. | ✗ |
| 0 | Use the pattern when a behavior makes sense only in some classes of a class hierarchy, but not in others. | ✗ |
| 2 | All mentioned | ✓ |

Which objects are involved in the Visitor pattern?



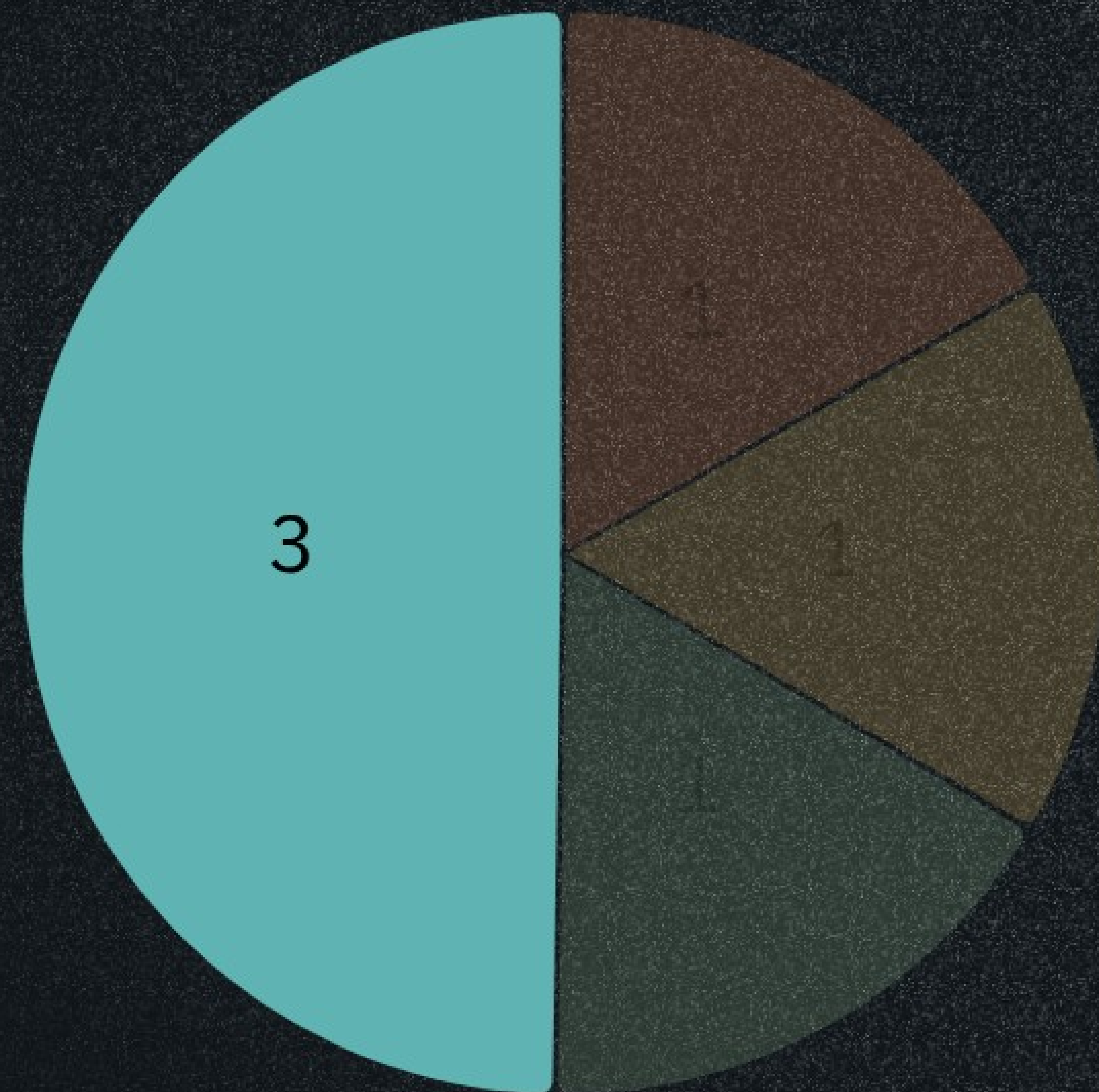
10 mins break

What is the primary purpose of the visitor pattern?



- ☐ 0 To enforce a singleton pattern ☐
- ☐ 0 To define a one-to-many dependency between objects ☐
- ☐ 0 To create families of related or dependent objects without specifying their concrete classes ☐
- ☒ 3 To define a new operation to be performed on each element in an object structure ☒
- ☐ 0 To provide a flexible way to create objects of different types ☐

What are the benefits of Visitor pattern?



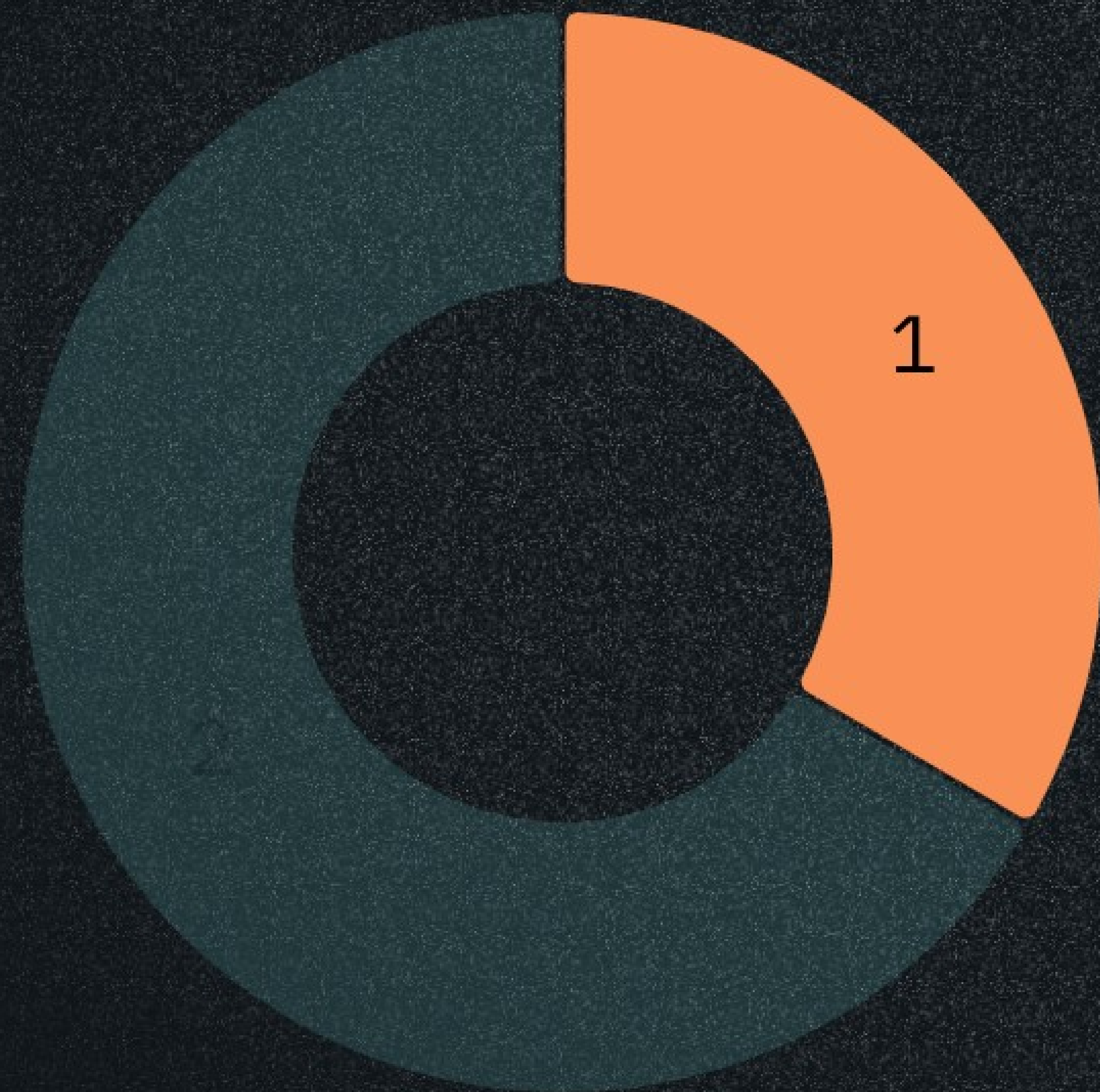
1 Open/Closed Principle. You can introduce a new behavior that can work with objects of different classes without changing these classes.

1 Single Responsibility Principle. You can move multiple versions of the same behavior into the same class.

1 Visitor can store some data while working with various objects. This can be handy when you want to traverse some complex object structure.

3 All mentioned

What are the drawbacks of the Visitor?



- ☒ 1 You need to update all visitors each time a class gets added to or removed from the element hierarchy. ✓
- ☒ 0 Visitors might lack the necessary access to the private fields and methods of the elements that they're supposed to work with. ✓
- ☐ 0 You can separate processing logic from data transfer objects. ✗
- ☐ 2 All mentioned ✗

Quiz

What does the Visitor pattern allow?

1 ✕



Direct modification of
objects

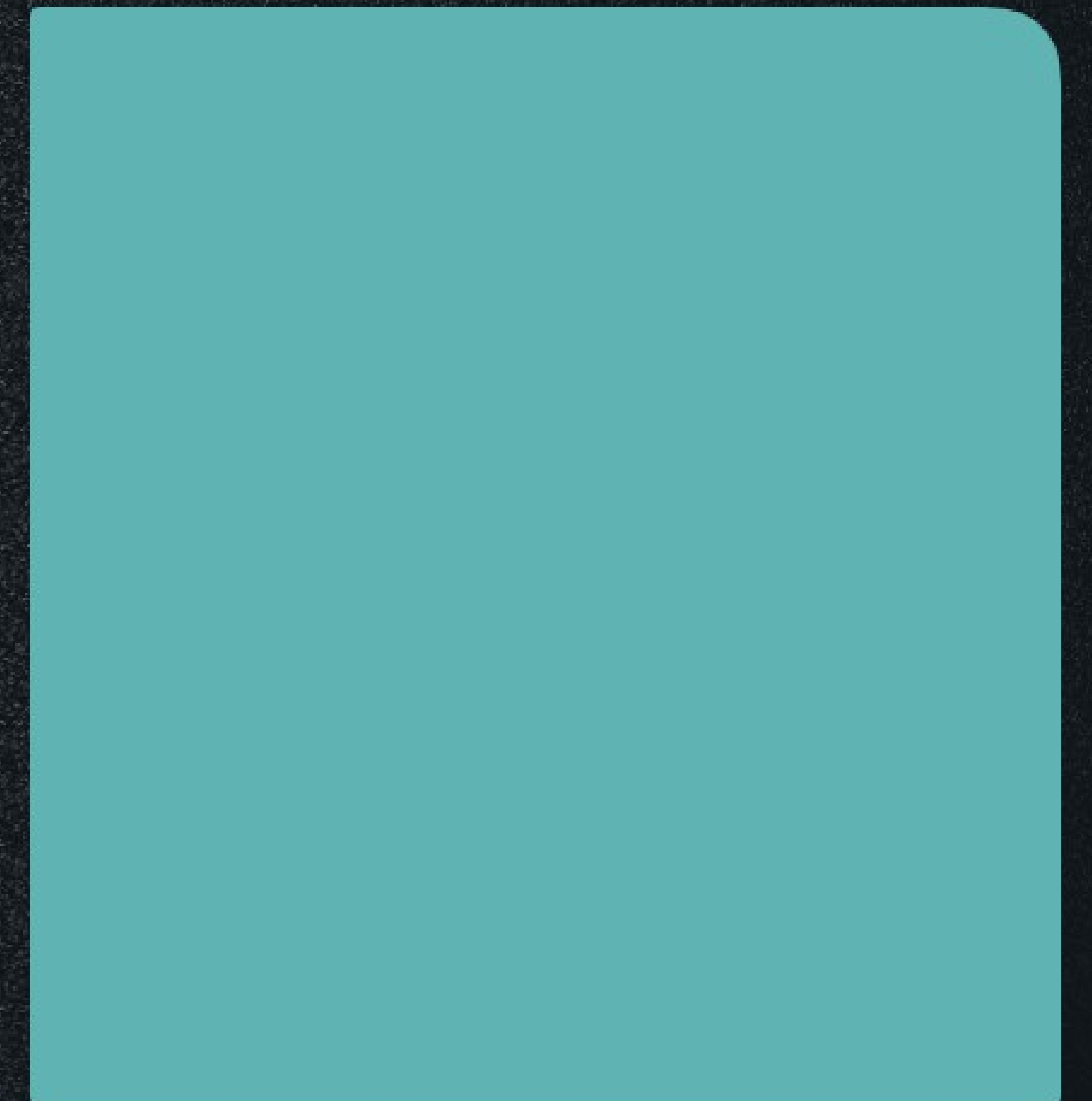
0 ✕

Simplified object
instantiation

0 ✕

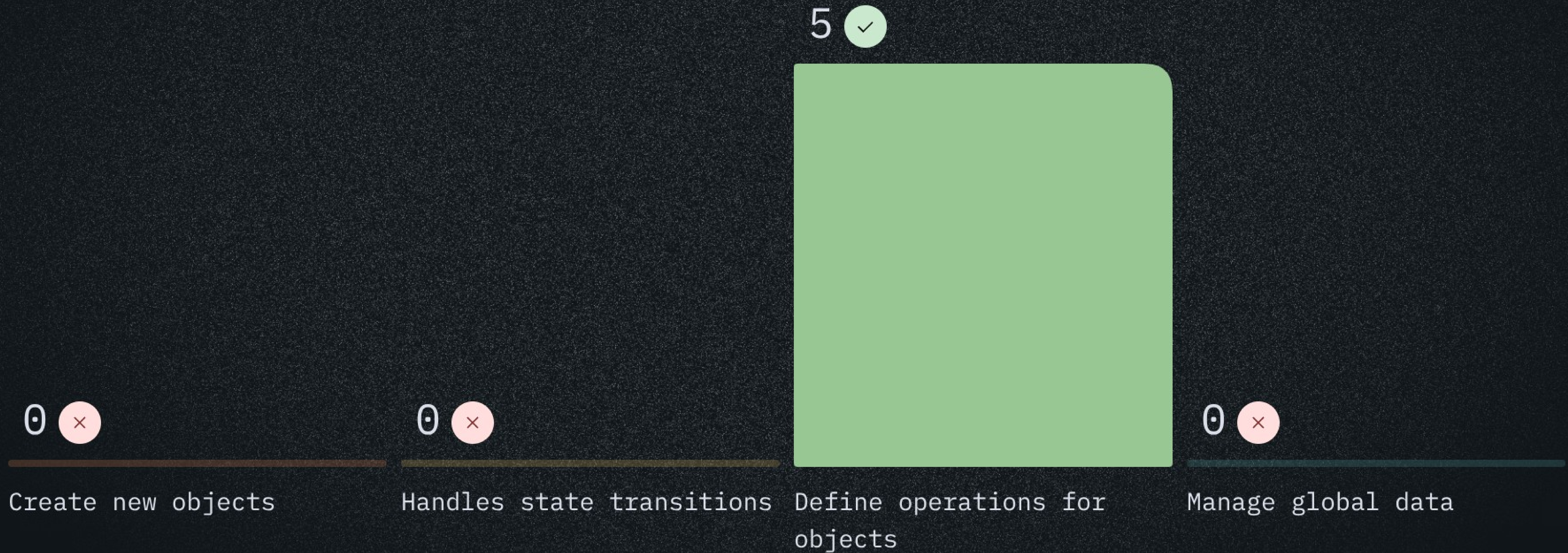
Object inheritance
injection

4 ✓



Adding new operations to
objects

What is the role of a Visitor class?



What does the accept method in Visitor do?

5 ✓



0 ✗

Declares factory level
classes

0 ✗

Initializes state of a
visitor

0 ✗

Adds observer state

When should the Visitor pattern be used?

1 ✕

For creating single objects

0 ✕

When objects rarely change

0 ✕

To remove simple loops

4 ✓

When operations on objects vary a lot

What does the Visitor pattern achieve?

5 ✓



0 ✗

Thread-safe objects

0 ✗

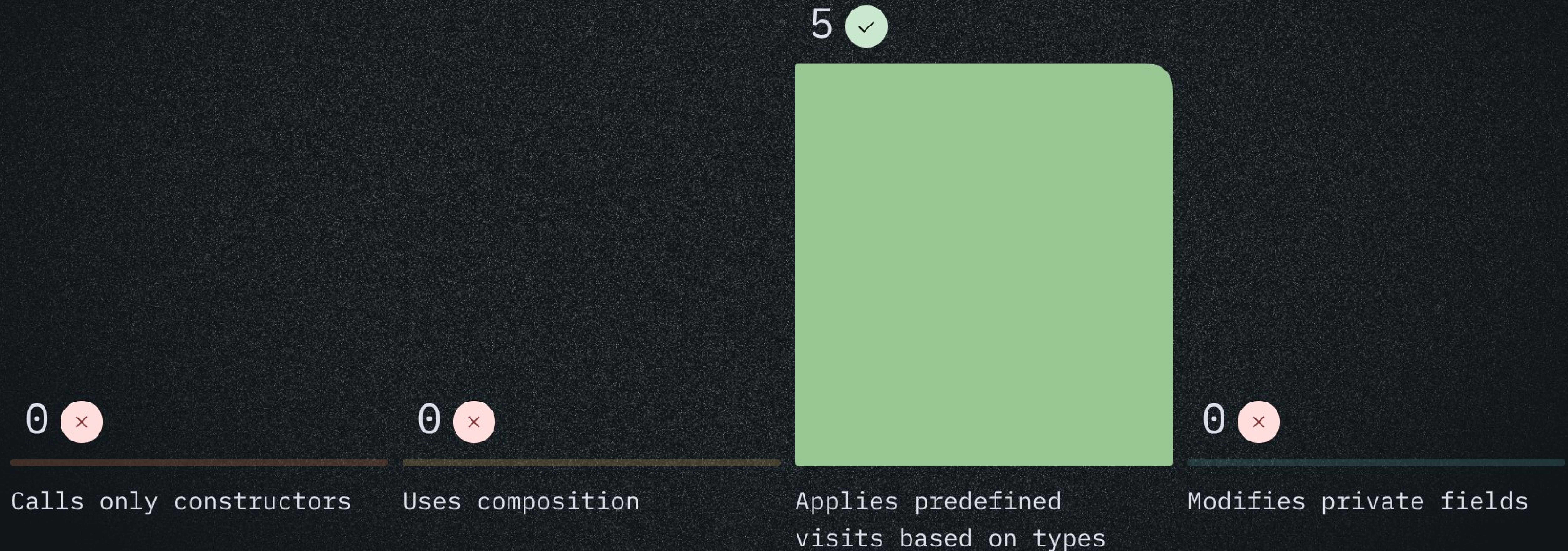
Multiple inheritance

0 ✗

Entity framework creation

Separation of object and operations

How does the Visitor interact with elements?



What is a drawback of the Visitor pattern?

2 ✓



Adding new objects is
difficult

0 ✗

It couples tightly to
factories

1 ✗



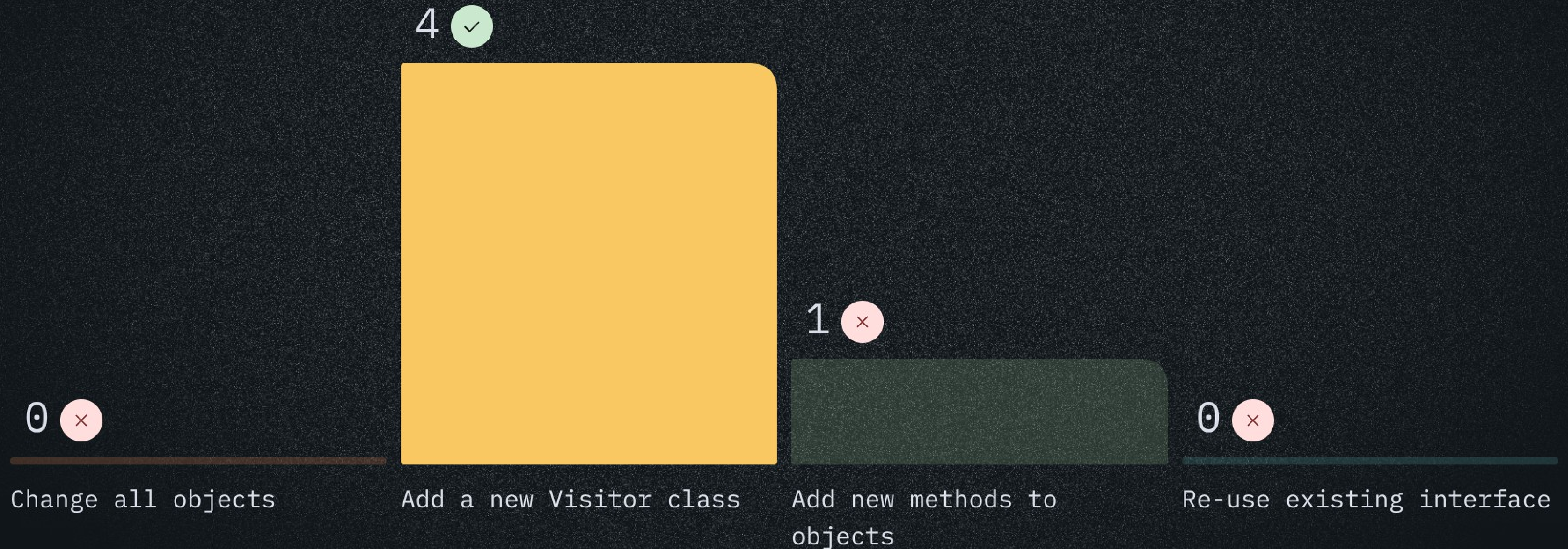
High memory consumption

2 ✗

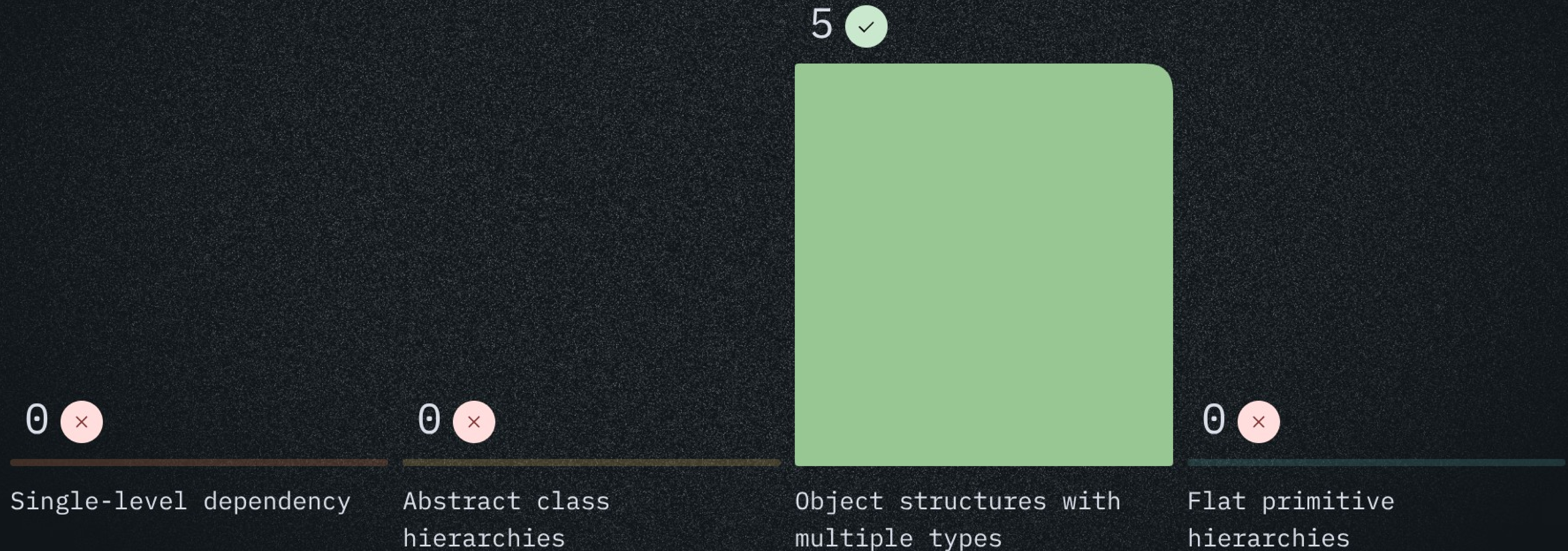


Cannot handle
polymorphic object trees

How does Visitor handle new operations?



What type of object relationships work best with Visitor?



Why use the Visitor pattern?

0 ✖

To remove complex data processing

0 ✖

To simplify object inheritance

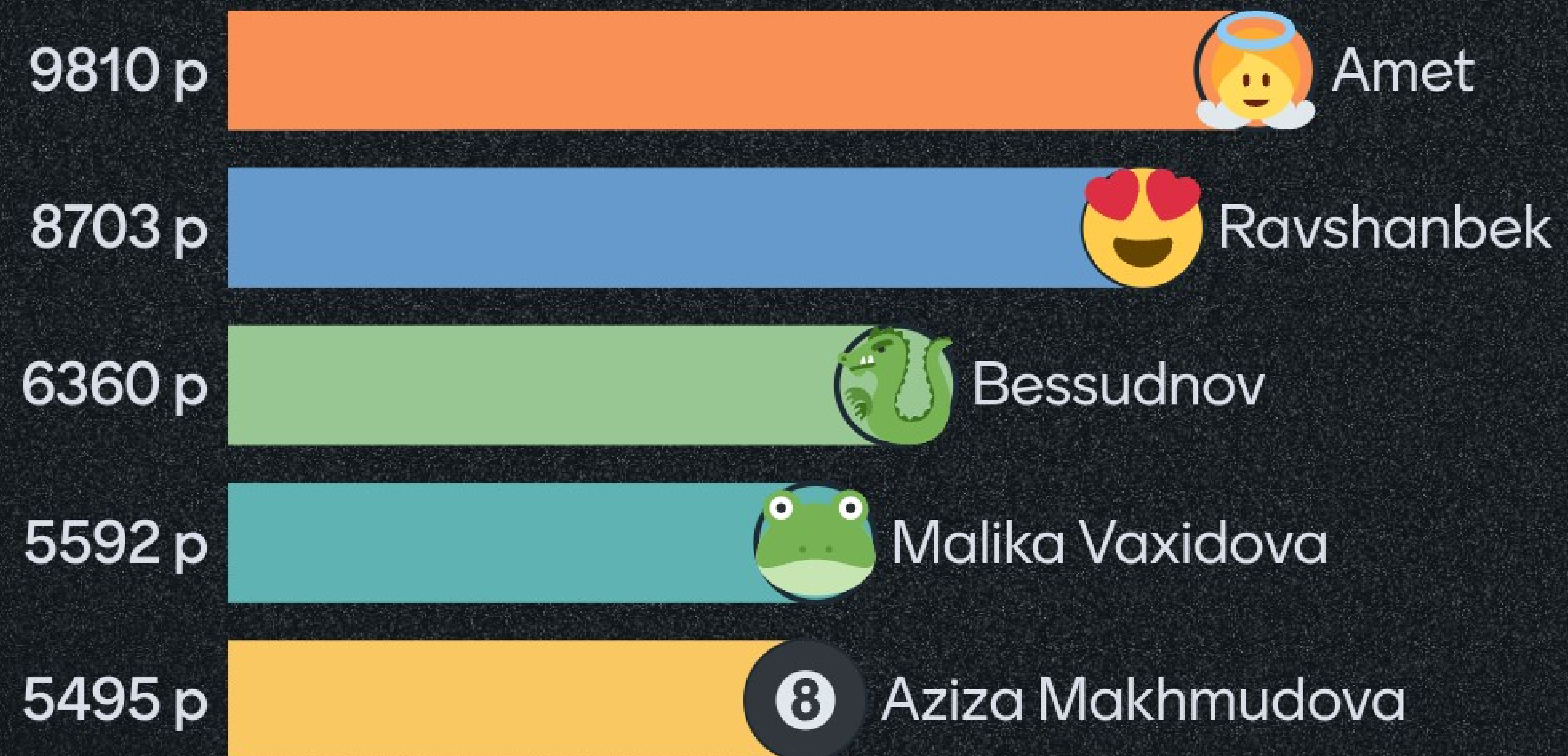
5 ✔

When adding behaviors without modifying classes

0 ✖

Ensure lightweight classes

Quiz leaderboard



Q&A part

0 questions

0 upvotes